

Why Auctions?

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JEB160 Auctions and Game Theory (Spring 2026)

I want to sell a good.

Price-maximizing seller

- Suppose you want to sell a unique object.
- Suppose you care only about the price.
- There is no 'organized market' for this specific object.
- Suppose potential buyers meet you randomly through an exogenous process, e.g., on a web platform.

Example: Selling a rare banknote



- a circulated banknote from 1911
- exceptional and premium paper quality (PMG certification: very fine 30 EPQ)

Price-maximizing seller

1. Posted price: a non-negotiable (take-it-or-leave-it/Buy-It-Now) price with immediate sale
 - selling to the first interested buyer upon her arrival
2. Preliminary price with an option to upgrade
 - collecting demands (waiting for the arrival of a sufficient number of potential buyers)
 - with a single buyer, immediate sale; with multiple buyers, soliciting competing bids/offers (auction)
3. Negotiable price (Accept Offers)
 - bargaining with buyers who give (counter)offers
4. Auction
 - a choice over several formats
 - possibly combined with a *reserve price* and/or *Buy-It-Now option*

Example: e-Bay auction for a rare banknote

Hi! [Sign in](#) or [register](#) Daily Deals Brand Outlet Gift Cards Help & Contact Ship to Sell Watchlist My eBay

ebay Shop by category Search for anything All Categories Search Advanced

LIVE Streaming now Shop exclusive items from trusted sellers Join event

22 VIEWED IN THE LAST 24 HOURS

PMG Dominion of Canada DC-18d-i 1911 \$1 - Series U-Y - Printer: ABNC S/N 975330-V pp D - Various | T.C. Boville - Black Line **30** Very Fine

ONE Dominion of Canada **ONE** **ONE DOLLAR**

True North Money (1368) **100% positive** · [Seller's other items](#) · [Contact seller](#)

C \$193.50
Approximately US \$140.37
18 bids · Ends in 1d 12h · Tuesday, 10:00 AM

Condition: --
"Very Fine 30 EPQ"

Place bid

[Add to Watchlist](#)

- 36 hours before: $193.50 \times 15.01 = 2904.43$ CZK
- final price: $510 \times 15.01 = 7665.10$ CZK

Why auctions?

- heterogeneous tastes (arts, antiques)
- rare (infrequent) transactions (both primary and secondary)
 - missing interbank markets (and little collateral) in transition countries; central bank organized auctions of credit
- rapid demand shifts (house sales with rapidly growing prices; past transactions are not a relevant benchmark)
 - 15-30% of property sales: HN August 22, 2025 There are “queues” for apartments. Interested parties compete to see who can offer the most. Price auctions are once again a common practice in sales.
- an organized secondary market doesn't exist for legal reasons
 - government licences, procurements

I want to 'do good' by selling goods.

Multi-objective seller

- revenues/price
- efficiency (the buyer has the highest valuation = no re-sale)
- equality/equity (e.g., think of 'voucher privatization' in transition economies, i.e., privatization *without* capital)
- transparency/credibility of the allocation mechanism (mainly to encourage participation)
- non-financial characteristics of buyers (when selling a house in a community, how do you find a good neighbor for your friends?)

Multi-objective seller: Posted price

- To ensure 'equity' among buyers, the seller may set a low price/fee.
- What could go wrong?
- Rationing due to excess demand
- Primary allocation through *queues* (or even corruption)
- Re-allocation through a *secondary market*

Example (Visapoint 2009+ at CZ embassies, esp. in Vietnam)

- queues for applications blocked by automatic software tools
- queue slots sold to applicants; the court decision is that it was legal

Fiasko kauzy Visapoint: stát musí vrátit zabavené stamiliony. A také platit odškodné



Marek Pokorný redaktor

9. 5. 2025 07:47 • Aktualizováno • 5 min. čtení ▶ PŘEHRÁT ČLÁNEK

ODEBRÁT AUTORA



Figure: HN May 9, 2025: Visapoint fiasco: the state must return the seized hundreds of millions. And also pay compensation

US electromagnetic spectrum

Centralized solutions

FCC (Federal Communications Commission) comparative hearings

- committees
- 1934-1981 (up to 1994): comparative hearings in US spectrum licences
- complex criteria (read 9 criteria in [wikipedia](#))
- administrative costs (bureaucracy)
- procedural details matter in committees (e.g., agenda-setting powers)
- information provision (lobbying of special interests)
- other pressure to influence committee membership

Centralized solutions

FCC comparative hearings

- legal battles with applicants
- sham applications (to profit from legal settlements)

4 • R&R August 27, 1999 **Radio Business**

Frozen Applicants Chilly To FCC Refund

☐ **Attorneys scramble to settle before auctions**

BY MATT SPANGLER
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Orion Communications has spent more than \$500,000 over the past 13 years in legal, engineering and other fees while waging a legal battle with the FCC to get WZLS-FM/Asheville, NC on the air. Last week the commission said Orion could have about \$2,500 of that money back.

That's how much Orion principal Betty Lee told R&R the broadcaster has spent on hearing fees. Those fees were paid to the commission in the days when it decided among several applicants who had applied for the same broadcast license through a series of lengthy and costly hearings.

Last week's announcement that the FCC was refunding the hearing fees in the four cases in which applicants had gone through at least part of a comparative hearing — only to have

APPLICANTS/See Page 10

- slow: many licenses unassigned

Decentralized solutions

Lotteries

- US Congress replaced the committee hearings on US spectrum licences with lotteries.
- 1981-1994: lotteries with no restriction on entry

Decentralized solutions

- many speculating applicants (400,000 applications)
 - ! e.g., a group of dentists won a license to run cellular phones on Cape Cod, then promptly sold it to Southwestern Bell for \$41 million
- inefficiencies: a low-valuation owner/speculator doesn't use the asset efficiently (e.g., low investments, unused licences, unused property)
- re-sales can fix inefficiencies of initial allocations, but involve delays and only pass on the original allocation problem to a new seller (with a different objective)
 - ! lotteries of small licenses contributed to the geographic fragmentation of the US cellular industry in the 1990s (in comparison to Europe)

Semi-centralized solutions

Auctions

- *decentralized* competition (bids) encourages *price discovery*
- *centralized* organization (auction rules, penalties for defaults, commitment of seller as the auctioneer is typically a platform) gives credibility
- designed to reach allocations faster than separate/bilateral trading
- extra criteria can be included (especially in multi-object auctions such as spectrum auctions)

I want to reduce 'liquidity stress'.

Covid-19 shock: March and April, 2020

- Covid-19 liquidity crisis/stress in Eurozone in Spring 2020
 - = market participants uncertain whether buyers of their papers exist; upward pressure on interest rates
- ECB established Pandemic Emergency Purchase Programme (PEPP) as a *quantitative easing* procedure
- PEPP was a *temporary asset purchase program* covering both private and *government* securities

Primary and secondary markets with government bills

Primary market (issuance) of the government securities

- organized through Treasury Bill auctions
- dealers buy bills from Treasury

Central bank purchases (PEEP)

= activities on the secondary markets

- organized either through bilateral purchases or **reverse Treasury auctions**
- dealers sell bills to the central bank
- each euro area central bank determines its own purchase procedure
- Do auctions on the secondary market affect the primary market differently than bilateral purchases?

Case study (Bats and Hoondert, 2022, ECB WP)

Eurozone offers a ‘policy experiment’

- treatment: auctions (for a part of allotments) in Netherlands (NL)
- controls: bilateral purchases in Germany and France (GER, FRA)
- ! euro area central banks operate under otherwise similar conditions (banking and monetary union) and are closely coordinated

B/C ratio (bid-to-cover) on the *primary* market is the main dependent variable

- = the total amount of bids over the total amount allotted
- measures the level of competition on the primary market
- a higher B/C ratio lowers not only primary but also secondary-market yields

Bid-to-cover ratio and allotments

Table 1: Descriptive statistics of the issuance data

Variables	Obs		Mean		Std Dev		Min		Max	
	Before	After	Before	After	Before	After	Before	After	Before	After
<i>Netherlands</i>										
B/C ratio	53	47	1.56	1.81	0.27	0.33	1.17	1.12	2.62	2.45
Allotted (bln)	53	47	1.53	1.55	0.78	0.74	0.40	0.40	3.09	2.99
<i>France</i>										
B/C ratio	210	105	3.40	2.96	1.05	0.78	1.42	1.33	8.06	5.24
Allotted (bln)	210	105	1.86	1.91	0.92	0.70	0.28	0.70	4.50	3.77
<i>Germany</i>										
B/C ratio	23	30	1.73	1.95	0.44	0.77	1.10	1.20	2.40	4.00
Allotted (bln)	23	30	1.55	1.54	0.69	0.66	0.40	0.40	2.79	2.80

Notes: This table presents the descriptive statistics of the b/c ratios and allotted amounts in the government auctions of the Dutch, French and German bill issuances separately. Each descriptive statistic is shown for the period before and after the start of the auctions by DNB.

Bid-to-cover ratio in Netherlands

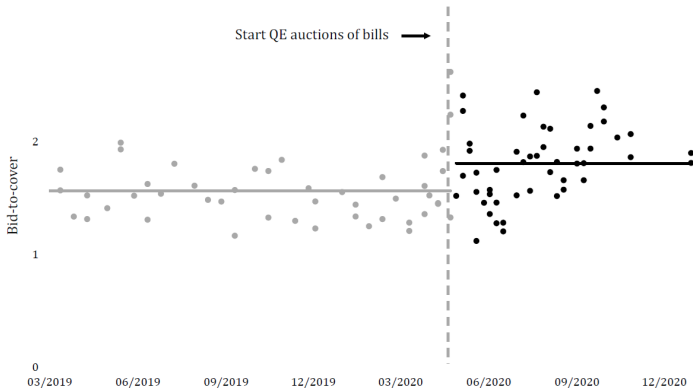


Figure 1: Bid-to-cover in the primary issuance of Dutch bills

Notes: This figure shows developments in the b/c ratio for the primary issuance of Dutch bills. The average b/c ratio is 1.56 before the start of the central bank auctions and 1.81 after the start of the central bank auctions.

Main analysis

$$Ratio_{i,c,t} = \alpha + \beta_1 \cdot Auction_{c,t} + \beta_2 \cdot Stress_{c,t} + \beta_3 \cdot Alloted_{c,t-1} + \varepsilon_{i,c,t}$$

- *Auction* = 1 for NL after April 2020; otherwise, *Auction* = 0
- *Stress* is a spread between the average yield and the 6-month OIS rate
- *Alloted* is a lagged allotment ($t - 1$ means 2nd + 3rd lag)
- i auction, c country, t date

Main analysis

Table 2: Central bank auctions effect on primary issuance bid-to-cover

Regressors	1	2
Auctions (dummy)	0.25*** (0.02)	0.23*** (0.02)
Market stress	-0.47** (0.24)	-0.39 (0.26)
Average allotted amount past issuance		-0.08** (0.03)
DE included in control group	Yes	Yes
FR included in control group	Yes	Yes
R-sqr	0.08	0.09
N (incl. issuance in DE and FR)	472	456

Notes: The dependent variable is the b/c ratio in the primary issuance auctions of bills. The data cover all Dutch, French and German primary issuances from January 2019 to January 2021 (at the transaction-level). To better compare the government auctions of bill issuance across jurisdictions, the b/c ratios are normalized on the basis of their country-specific average during the period before the start of the DNB auctions. Constant not shown. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Extra analyses

- B/C Ratio on the primary market: when re-estimated with a dummy for Stress (spread above 75th percentile), the base increase is 20%; in times of market stress, it is 26%
- Offers in reverse Treasury auctions (secondary market): around primary issuance dates, the offers are larger (in size) and more competitive (lower yields)

Reverse Treasury auctions during a liquidity crisis

- + public announcement of credible implementation (esp. timing and scale) helps to avoid 'loss of confidence'
- + published aggregated data on the allocated offer prices and sizes allow for 'price discovery'
- This example shows that central banks serve as *market makers of last resort* and restore bill market functioning during a liquidity crisis, especially via auctions.

US food banks

Feeding America

Priority Allocation System

- 1976-2005: 'wait your turn' system
 - each food bank has a *position* in the queue
 - position set by (i) *quantity* of the poor (3 weighted groups defined relative to the poverty line) and (ii) *recent allocations* to the food bank
 - truck loads offered sequentially to the queue
 - product mix in a truck more or less similar, but food banks pay transportation costs
 - food bank could reject an offer (within a few hours) and the offer is made to another bank
 - with acceptance, its position drops recalculated; see criterion (ii)
 - with rejection, its position almost unaffected
- + simple
- + equity: treats poor from all food banks equally

Feeding America

Priority Allocation System

- ! 75% of food delivery is decentralized, not through the centralized system; consequently demand is largely *heterogenous*
 - poor are not served equally
 - large differences in product inventories
 - also different storage capacities, especially refrigerators' capacities
- priority didn't reflect actual demand = rejections
- delays lead to spoilage (esp. fresh food close to its expiry)
- delays costly for donors (cannot wait for several banks to reply)

Feeding America

Choice System: Auctions

- virtual money: *shares*
- shares used by food banks two *auctions* per day
- first-price *sealed-bid* auctions (to avoid 'sniping' by more sophisticated food banks)
- fractional bidding: small banks can bid jointly
- negative bids allowed: this addresses that donors need to remove food quickly
- food banks can also auction off food from other sources
- shares cannot be traded for anything other within the auction market

Feeding America

Choice System: Budgets

- initial amounts of *shares* reflect the *quantity* of the poor
- recirculation: all shares used in the day's auctions are pooled and then redistributed overnight
- redistribution to winners reflects the (relative) quantity of the poor served by wins
- short-term *credit* allowed for *small* banks to increase temporarily insufficient balances (used increasingly over time)
- money supply: with an increase of total supply of food (more truckloads), the share balances are adjusted to keep prices stable

Feeding America: Choice System



You needn't be a food-insecure person to have some familiarity with the United States' network of organizations devoted to fighting hunger. You may have volunteered at one of the country's 60,000 food pantries and meal programs. Or donated to one of its 200 food banks. Or know one of the 10 million–12 million Americans who visit a food pantry in a given year. This network is as vast as it is

Figure: A short summary available in the [Chicago Booth Review](#)

Toolkit: Game theory

Essentials

- Non-cooperative game theory is ‘the language’ of modern microeconomics.
- The idea is to model rational players who optimize against each other.
 - 1-person decision-making theory
 - n -person decision-making theory, $n \geq 2$
- Why (and when) rationality?
 - Strong incentives to correct errors in high-stakes decisions.
 - Experience accumulated in frequently repeated interactions.
 - Survival selection and imitation of successful strategies.
- More on rationality in the next lecture.
- Main alternatives to ‘classic game theory’:
 - behavioral game theory: systematic violations to optimization, e.g., due to cognitive costs
 - evolutionary game theory: strategies chosen by heuristics (imitation) and by survival pressure (selection)
 - algorithmic game theory: optimization reflects computational constraints (machines)

A brief history

1930s Applied math

1940s Zero-sum games during the World War 2

1950s Deterrence to analyze the Cold War

1960s Threats and bargaining (industrial relations, politics, international relations)

1970s Evolutionary biology

1990s Auctions

2000s Market design

2010s Networks, communication

2020s Algorithms, machines

How about (game/microeconomic) theory and empirics?

- *Experiments*: alternative mechanisms/games directly tested in the lab or field
- *Market (and organization) design*: alternative mechanisms proposed, developed, and tested (auctions, matching)
- *Reduced-form empirics*: microeconomic models used to guide the estimation and explain the signs of estimates (qualitative predictions)
- *Structural-form empirics*: microeconomic models to 'closely' fit the problem, derive (quantitative) parameters, and then to serve for analysis of structural and policy changes
- *Business strategy*: models and experience to guide managerial practices

Course organization

- classes are voluntary but strongly recommended
- slides online, often later slightly modified/improved
- we'll play a few auctions
- homeworks individually or in couples
- homeworks take time, but these investments pay off!
- a written closed-book exam (sample exam is online)
- the exam is based on the problem sets in the class and homeworks

Enjoy the course and good luck!

References

- Bats, J. V., Hoondert, J. J. (2022). The relationship between central bank auctions and bill market liquidity. ECB Working Paper 2708.
- Prendergast, C. (2017). How food banks use markets to feed the poor. *Journal of Economic Perspectives*, 31(4), 145–162.